

SDV_Replica_POC

Daily Work Log

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Project: SDV_Replica_POC

Log version: v3.0

Focus of this session: fix a repo-wide CRLF line-ending issue found in the uploaded working copy; add real, verified C++ build scaffolding for all 4 HPC services; build a cmd.exe-native, menu-driven build console covering every module in the repo; and update documentation (README, CHANGELOG, per-module READMEs) to match. Repo tag going into this session: v0.2.0.

1. REPO HEALTH CHECK - CRLF LINE-ENDING ISSUE

Item	Result
Uploaded repo showed 46 files as 'modified' vs. last commit (v0.2.0)	INFO
Diffed with whitespace ignored (git diff -w) - zero real content differences	OK
Confirmed root cause: Windows checkout converting LF -> CRLF on nearly every text file	INFO
scripts/buildenv.sh specifically confirmed to have CRLF line terminators (file command)	FAIL
Discarded the CRLF-only working-tree changes (git checkout -- .)	OK
.gitattributes hardened: explicit 'text eol=lf' rules per extension, .bat kept as eol=crlf	OK

In plain terms: A bash script with CRLF line endings can fail outright on Linux/Git-Bash with a 'bad interpreter' error, even though it looks completely normal when opened in a Windows editor. The previous .gitattributes only marked binary files (PDFs, images) - it never told Git how to handle line endings on text files, so Windows' default checkout behavior was free to silently convert them. The fix makes Git enforce LF on every source/script file regardless of which OS checks it out, so this cannot recur silently again.

2. C++ BUILD SCAFFOLDING - 4 HPC SERVICES

Added CMakeLists.txt + src/main.cpp for all 4 HPC services referenced in the architecture doc, matching the exact folder paths already in the repo:

- hpc1-rpi5/dom0-services/hpc-bridge — HPC-1 to HPC-2, over the Ethernet backbone
- hpc1-rpi5/dom0-services/cloud-gateway — HPC-1 to AWS IoT Core
- hpc2-bbb/containers/hpc-bridge — HPC-2 to HPC-1
- hpc2-bbb/containers/cloud-gateway — HPC-2 to Azure IoT Hub

Item	Result
hpc1-rpi5/dom0-services/hpc-bridge — cmake configure + build	OK

hpc1-rpi5/dom0-services/cloud-gateway — cmake configure + build	OK
hpc2-bbb/containers/hpc-bridge — cmake configure + build	OK
hpc2-bbb/containers/cloud-gateway — cmake configure + build	OK
All 4 binaries executed directly - each printed its own distinct identity string	OK

SCOPE - what this is and is not

This is a HOST-MACHINE sanity build (C++17, g++, CMake) - it proves the build structure and catches syntax errors early. It is NOT yet cross-compiled for the RPi5 (aarch64) or BBB (armhf) target architectures, and none of the 4 services have real gRPC/protobuf logic wired in yet - every main.cpp has explicit TODO comments marking exactly what to add next and where.

3. CMD.EXE BUILD CONSOLE - scripts/build_console.bat

Built a menu-driven build console, styled after a reference example the project owner shared (a build-menu CLI from an unrelated project), but wired to this repo's own real modules - not the reference example's modules, which belong to a different, unrelated project (different boards, different AUTOSAR-based structure).

Item	Result
Menu: clean/incremental x (all / ZCU-only / HPC-only / specific module)	OK
Per-module: source file count, live [COMPILING...] status, PASS/FAIL verdict	OK
Per-module timestamped log under Logs/Build//	OK
Session summary (BUILD SESSION VERDICT) at the end of each menu action	OK
Individual commands (cmake -S / cmake --build) verified separately	OK
Full end-to-end run on real Windows cmd.exe	PENDING

HONESTY NOTE - this file is not yet proven end-to-end

Every command inside build_console.bat was verified individually (the same cmake configure/build steps used successfully for both the ZCU firmware and the new C++ services). The .bat file as a WHOLE has not had a real run on Windows cmd.exe, because the environment that wrote it has no Windows machine to test on. This is different from every other script in this repo, which was run and its output captured before being handed over. Run it and report the first thing that looks wrong, the same way earlier issues in this project were debugged.

4. DOCUMENTATION UPDATES

- CHANGELOG.md — added v0.3.0 entry covering all of the above.
- README.md — version bumped to v0.3.0, added a Build Tooling section documenting both scripts/buildenv.sh and scripts/build_console.bat, updated Status section.
- All 4 HPC service READMEs — version bumped, noted the new build scaffolding and what's still a TODO in each.

- scripts/README.md — added full build_console.bat documentation alongside the existing buildenv.sh docs.

5. NEXT STEPS

- Run scripts/build_console.bat on real Windows and report the first issue, if any.
- Merge these changes into the working repo (git add / commit / tag v0.3.0 / push).
- STM32 Discovery/Nucleo exact part numbers still not confirmed (Ethernet PHY question).
- Generic linker script / vector table still not matched to real datasheet values.
- RPi5 Stage 1 hardware bring-up still not started.
- gRPC/protobuf logic and mTLS 1.3 credentials still TODO in all 4 HPC service main.cpp files.

End of log - 2026-08-30. Repo: github.com/SJLNS/SDV_Replica_POC